STATISTICAL BUSINESS ANALYSIS REPORT

1. Executive Summary:
Statistical analysis conducted on sales and customer churn datasets to evaluate business revenue drivers and retention factors.

2. Descriptive Statistics & Confidence Intervals:
- Mean Total Sales: $123,650.48 ± $19,874.13 (95% CI)
- Median Sales: $83,758.00 | Standard Deviation: $100,125.04

3. Hypothesis Testing:
- **Two-Sample t-Test:** Significant difference in monthly charges between churned and retained customers (t = -2.4102, p = 0.0163). Higher charges correlate with attrition.
- **One-Way ANOVA:** Regional variations in sales volume are not statistically significant (F = 2.1644, p = 0.0972).
- **Chi-Square Test:** Strong relationship between contract type and churn (Chi2 = 27.7153, p < 0.001). Month-to-month contracts exhibit highest churn.

4. Linear Regression Analysis:
- Model: Total Sales = -121,700 + 26,430(Quantity) + 4.61(Price)
- R-squared: 0.884 (88.4% variance explained by order quantity and unit price).

5. Business Recommendations:
- Promote annual/two-year contracts to curb customer churn.
- Standardize cross-regional operational efforts as baseline regional variance is non-significant.